Linear Regression

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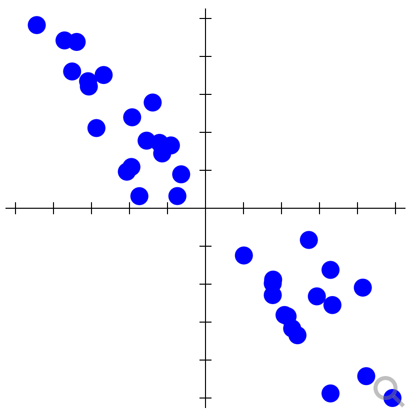
Question 1

0/1 pt ⟲ 3 ⇄ 19

Here is a scatter plot for a set of bivariate data.



What would you estimate the correlation coefficient to be?

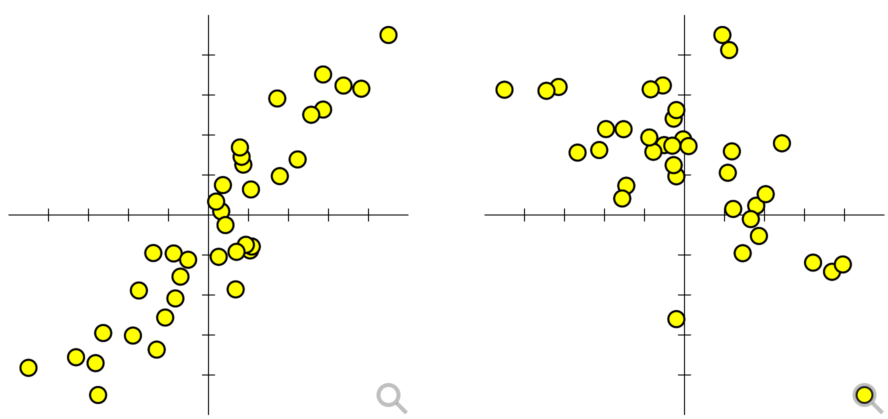
-0.9 -0.6 0 0.6 0.9

Question Help: Video

Question 2

0/1 pt ⟲ 3 ⇄ 19

Here are the scatter plots for two sets of bivariate data with the same response variable. The first compares the variables *x* & *y*. The second compares the variables *w* & *y*



Which explanatory variable has a stronger relationship with the response variable *y*?

- The first variable (*x*) has a stronger relationship with the response variable *y*.
- The second variable (*w*) has a stronger relationship with the response variable *y*.

Question Help: Video

Question 3

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Based on the data shown below, a statistician calculates a linear model $y = -2.53x + 37.56$.

x	y
3	30.9
4	29.38
5	24.06
6	20.84
7	18.72
8	16.3
9	15.88
10	11.96
11	9.54
12	8.22
13	5.1

Use the model to estimate the y-value when $x = 3$

y = ______

Use the model to estimate the y-value when $x = 4$

y = ______

Question Help: Video

Question 4

0/1 pt 3 19

A regression was run to determine if there is a relationship between hours of TV watched per day (x) and number of situps a person can do (y).

The results of the regression were:

```
y=a+bx
b=-1.004
a=32.933
r²=0.561001
r=-0.749
```

Use this to predict the number of situps a person who watches 3.5 hours of TV can do.

______ Round to one decimal place.

Question Help: Video

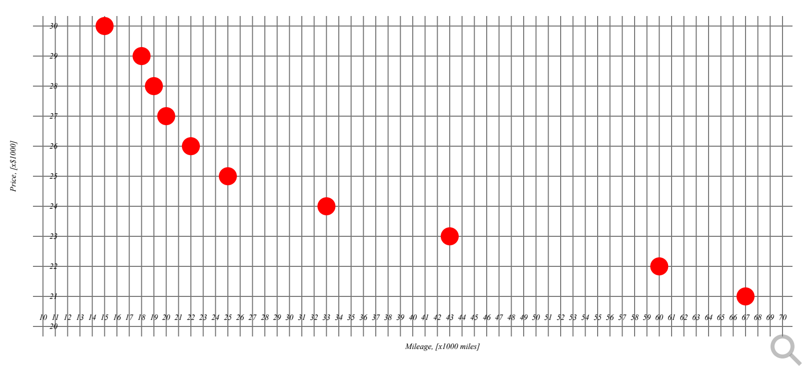
Question 5

0/1 pt 3 19

Consider the following data set that contains information about a sample of ten 2014 Honda Accord offered for sale.

Vehicle	Condition	Color	Mileage (x1000miles)	Price (x$1000)
1	excellent	blue	15	30
2	average	blue	18	29
3	poor	blue	19	28
4	poor	black	20	27
5	average	green	22	26
6	poor	silver	25	25
7	poor	red	33	24
8	good	silver	43	23
9	average	blue	60	22
10	poor	black	67	21

The scatter plot that summarizes the data with regard to mileage as the input variable and price as the output variable is provided below.



a. Is there an association between the two variables? If yes, is it positive or negative?

There appears to be [Select an answer] between the two variables.

b. Use technology to find the equation of the line of best fit, $y = mx + b$, and the coefficient of determination and interpret the slope, the vertical intercept, and the coefficient of determination. (Round the answers to 3 decimal places.)

i. $m =$ ______ [Select an answer] is the [Select an answer] ;

ii. $b =$ ______ [Select an answer] is the [Select an answer] ;

iii. $r^2 =$ ______ hence there is a [Select an answer] linear relation.

c. Use the line of best fit to:

i. estimate the price of a car with 43500 miles.

Answer: ________ dollars (Round the answer to the nearest unit.)

ii. estimate the mileage of a car whose price is 26500 dollars.

Answer: ________ miles (Round the answer to the nearest unit.)

Question Help: Video

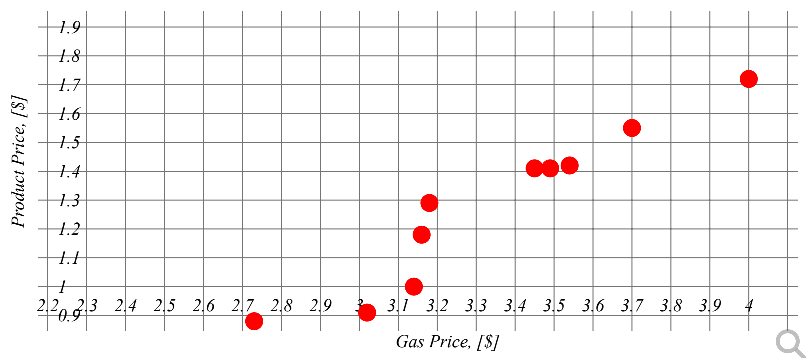
Question 6

0/1 pt 3 19

Consider the following data set that contains information about the average price of a dozen eggs in ten randomly chosen cities:

City	Gas price per gallon ($)	Price of a dozen eggs ($)
Salt Lake City	2.73	0.88
Miami	3.02	0.91
San Antonio	3.14	1
Pittsburgh	3.16	1.18
Denver	3.18	1.29
Tampa	3.45	1.41
New Orleans	3.49	1.41
San Francisco	3.54	1.42
Phoenix	3.7	1.55
Orlando	4	1.72

The scatter plot that summarizes the data with regard to price of gas as the input variable and price of a dozen eggs as the output variable is provided below.



a. Is there an association between the two variables? If yes, is it positive or negative?

There appears to be a [Select an answer] association between the two variables.

b. Use technology to find the equation of the line of best fit, $y = mx + b$, and the coefficient of determination and interpret the slope, the vertical intercept, and the coefficient of determination. (Round the answers to 3 decimal places.)

i. $m =$ ______ [Select an answer] is the [Select an answer] ;

ii. $b =$ ______ [Select an answer] is the [Select an answer] ;

iii. $r^2 =$ ______ hence there is a [Select an answer] linear relation.

c. Use the line of best fit to:

i. estimate the price of a dozen eggs when the price of gas is 2.45 dollars per gallon.

Answer: ______ dollars. (Round the answer to 2 decimal places)

ii. estimate the price of gas when the price of dozen eggs is 1.03 dollars.

Answer: ______ dollars per gallon. (Round the answer to 2 decimal places)

Question Help: Video

Question 7

0/1 pt 3 19

Use the equation $J = 179W - 937.7$ to

i. find the output J when the input $W = -2.6$.

Answer: ______ (Round the answer to 2 decimal places)

ii. find the input W when the output $J = -120.8$.

Answer: ______ (Round the answer to 2 decimal places)

Question Help: Video

Question 8

0/1 pt 3 19

Fill out the table: (Round the answers to 2 decimal places)

x	y	$x-\bar{x}$	$(x-\bar{x})^2$	$y-\bar{y}$	$(y-\bar{y})^2$	$(x-\bar{x})(y-\bar{y})$
2	0	-0.5	0.25	-2	4	1
3	4	______	______	______	______	______
$\bar{x} =$ ______	$\bar{y} =$ ______	$S_{xx} =$	______	$S_{yy} =$	______	$S_{xy} =$ ______

Question Help: Video 1 Video 2

Question 9

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x	y	$x-\bar{x}$	$(x-\bar{x})^2$	$y-\bar{y}$	$(y-\bar{y})^2$	$(x-\bar{x})(y-\bar{y})$
0	0	-2	4	-1	1	2
4	2	2	4	1	1	2
2	1	2	4	1	1	2
$\bar{x} =$ 2	$\bar{y} =$ 1	$S_{xx} =$	12	$S_{yy} =$	3	$S_{xy} =$ 6

Use the results from the table above to compute the slope and the y-intercept of the regression line:

$$m = \frac{S_{xy}}{S_{xx}} = \underline{\qquad}$$ (Round the answer to 2 decimal places)

$$b = \bar{y} - m\bar{x} = \underline{\qquad}$$ (Round the answer to 2 decimal places)

Question Help: Video

Question 10

0/1 pt 3 19

x	y	$x-\bar{x}$	$(x-\bar{x})^2$	$y-\bar{y}$	$(y-\bar{y})^2$	$(x-\bar{x})(y-\bar{y})$
0	1	-1	1	-1.33	1.77	1.33
1	2	0	0	-0.33	0.11	-0
2	4	1	1	1.67	2.79	1.67
$\bar{x} =$ 1	$\bar{y} =$ 2.33	$S_{xx} =$	2	$S_{yy} =$	4.67	$S_{xy} =$ 3

$$y = (1.5)x + (0.83)$$

Use the results from the table above to compute the coefficient of determination:

$$r^2 = \frac{S_{xy}^2}{S_{xx}S_{yy}} = \underline{\qquad}$$ (Round the answer to 4 decimal places)

Question Help: Video

Question 11

0/1 pt 3 19

x	y	$x-\bar{x}$	$(x-\bar{x})^2$	$y-\bar{y}$	$(y-\bar{y})^2$	$(x-\bar{x})(y-\bar{y})$
2	4	-0.33	0.11	2	4	-0.66
4	0	1.67	2.79	-2	4	-3.34
1	2	-1.33	1.77	0	0	-0
$\bar{x} =$ 2.33	$\bar{y} =$ 2	$S_{xx} =$	4.67	$S_{yy} =$	8	$S_{xy} =$ -4

$$y = (-0.86)x + (4),\, r^2 = 0.4283$$

Use the results above to compute the linear correlation coefficient:

$r =$ ______ (Round the answer to 4 decimal places)

Question Help: Video